Figure 9.3 shows the construction of an example tree for the unidimensional case. This is an edge case, because singletons (tuples with dimension 1) will result in always using the x coordinate in points (i.e. the whole singleton).

Notice how each node of the tree “covers” a region of the dataset, in a hierarchical way: the root covers the whole dataset, level 1 nodes cover the left and right partitions created using the root as a pivot, and level 2 nodes cover the even smaller partitions created using level 1 nodes as pivots.

Here, when we use the term “cover”, we mean that, given an entry X in the search space (in 1-D, given a real number X), and if we query the tree (as a binary search tree) for X, then all nodes we traverse during the search for X (which form a path from the root to a node N) cover X. In particular, the node on which search stops is the one covering the smallest region for X.

In other words, each node in the tree has associated a range of values for which it will be traversed, when searching the tree for those values; that range is the region covered by the node.

Make sure to go through the example in figure 9.3 and to understand every step, maybe even try to run an example yourself (for instance by just changing the pivots or their order, and checking how the tree changes). It is important understanding the unidimensional case, because it will make it simpler to understand the 2d-tree construction.